# MATH 1400 · Quiz 5 · Side 1 — Arc length · Surface area · Average value · Centroids

Sections in solve order: pick the tool → set up → see/do → finish → check

## 0 · Which tool?

| The question says… | Use |
|---|---|
| "length of the curve" | $L = \int dL$ |
| "area of the surface" rotated about an axis | $S = \int 2\pi r \, dL$ |
| "average value of $f$ on $[a,b]$" | $\frac{1}{b-a}\int_a^b f(x)\,dx$ |
| "average of $g$ along a curve / on a surface" | $\frac{\int g\,dL}{\int dL}$, $\frac{\int g\,dS}{\int dS}$ |
| "centroid" (no density) / "center of mass" ($\rho$ given) | moments ÷ area (or mass), §4 |
| "set up, do not evaluate" | full integral with bounds and $dx$, then stop |

## 1 · Arc length

$$L = \int_a^b \sqrt{1 + \left(\tfrac{dy}{dx}\right)^2}\,dx \quad \text{(curve given as } y = f(x)\text{)}$$
$$L = \int_c^d \sqrt{1 + \left(\tfrac{dx}{dy}\right)^2}\,dy \quad \text{(curve given as } x = g(y)\text{)}$$
$$L = \int_\alpha^\beta \sqrt{\left(\tfrac{dx}{dt}\right)^2 + \left(\tfrac{dy}{dt}\right)^2}\,dt \quad \text{(parametric)}$$

**Key move:** differentiate → square → add 1 → find the perfect square → √ cancels.

| See this | Do this |
|---|---|
| $y' = A - B$ with $A \cdot B = \frac{1}{4}$: $\frac{x^3}{6} + \frac{1}{2x}$, $\frac{x^4}{16} + \frac{1}{2x^2}$, $x^2 - \frac{\ln x}{8}$, $\frac{2}{3}x^{3/2} - \frac{1}{2}x^{1/2}$, $\frac{1}{4}y^{-2} + \frac{1}{8}y^4$, $\frac{e^x + e^{-x}}{2}$ | $1 + y'^2 = (A+B)^2 \Rightarrow$ integrand is $A + B$ (flip the middle sign) |
| $y = c\,x^{3/2}$ or $c(\dots)^{3/2}$ | $y'^2$ is a polynomial. $\frac{4}{3}x^{3/2}$: $\sqrt{1+4x}$, u-sub. $\frac{2}{3}(4x^2+\frac{1}{2})^{3/2}$: $1 + y'^2 = (16x^2+1)^2$ |
| $y = \ln\cos x$ | $1 + \tan^2 x = \sec^2 x \Rightarrow dL = \sec x\,dx$ |
| $y = \ln(1 - x^2)$ | $\sqrt{} = \frac{1+x^2}{1-x^2} = -1 + \frac{1}{1-x} + \frac{1}{1+x}$ |
| $y = \cosh x$ or $x = \cosh y$ | $1 + \sinh^2 = \cosh^2 \Rightarrow dL = \cosh\,dx$ |
| $y = \int_0^x g(t)\,dt$ | FTC: $y' = g(x)$. $\sqrt{e^{2x} - 1} \Rightarrow dL = e^x\,dx$ |
| $y = \frac{3}{2}x^{2/3}$ | $\sqrt{} = \frac{\sqrt{x^{2/3}+1}}{x^{1/3}}$, $u = x^{2/3} + 1$ |
| Parametric powers $x = at^m$, $y = bt^n$ | Factor out the smallest power: $\sqrt{24^2 t^{14} + 24^2 t^{22}} = 24t^7\sqrt{1+t^8}$, $u = 1 + t^8$ |
| Parametric polynomials | $x'^2 + y'^2$ is a square: $(2t)^2 + (t^2-1)^2 = (t^2+1)^2$ |

**Example.** $y = \frac{x^3}{6} + \frac{1}{2x}$, $1 \le x \le 3$: $y' = \frac{x^2}{2} - \frac{1}{2x^2}$, $1 + y'^2 = \left(\frac{x^2}{2} + \frac{1}{2x^2}\right)^2$, $L = \int_1^3 \left(\frac{x^2}{2} + \frac{1}{2x^2}\right)dx = \frac{14}{3}$.

## 2 · Surface area of revolution

$S = \int 2\pi\, r\, dL$. About the **x-axis**: $r = y$. About the **y-axis**: $r = x$.

**Use $dL$, never plain $dx$.** Write $r$ in the integration variable.

| Given | about x-axis | about y-axis |
|---|---|---|
| $y = f(x)$, $a \le x \le b$ | $\int_a^b 2\pi f(x)\sqrt{1 + f'^2}\,dx$ | $\int_a^b 2\pi x\sqrt{1 + f'^2}\,dx$ |
| $x = g(y)$, $c \le y \le d$ | $\int_c^d 2\pi y\sqrt{1 + g'^2}\,dy$ | $\int_c^d 2\pi g(y)\sqrt{1 + g'^2}\,dy$ |
| $x(t), y(t)$ | $\int 2\pi y(t)\sqrt{x'^2 + y'^2}\,dt$ | $\int 2\pi x(t)\sqrt{x'^2 + y'^2}\,dt$ |

| See this | Do this |
|---|---|
| $y = \sqrt{x}$ about x-axis | $r\,dL = \sqrt{x}\cdot\frac{\sqrt{4x+1}}{2\sqrt{x}}dx = \frac{1}{2}\sqrt{4x+1}\,dx$ |
| $y = \sqrt{R^2 - x^2}$ or $x = \sqrt{R^2 - y^2}$ (circle) | $r\,dL = R\,dx \Rightarrow S = 2\pi R\cdot$ (width). Sphere $= 4\pi R^2$ |
| $x\sqrt{1+4x^2}$, $x^3\sqrt{1+x^4}$, $y^3\sqrt{1+9y^4}$ | u = what's under the root; odd bounds like $\sqrt[4]{8}$ make $u$ a perfect square |
| $x = \frac{1}{3}(y^2+2)^{3/2}$ | $1 + x'^2 = (y^2+1)^2 \Rightarrow dL = (y^2+1)\,dy$ |
| $y\sqrt{1+y}$ after $u = 1 + y$ | replace $y = u - 1$ |
| "first quadrant", e.g. $y = 9 - x^2$ | bound = where curve hits the axis ($x = 3$) |

**Example.** $y = \sqrt{x}$, $0 \le x \le 2$, about x-axis: $S = \pi\int_0^2 \sqrt{4x+1}\,dx = \frac{\pi}{6}(27 - 1) = \frac{13\pi}{3}$.

## 3 · Average value

$f_{avg} = \frac{1}{b-a}\int_a^b f(x)\,dx$. MVT: some $c \in (a,b)$ has $f(c) = f_{avg}$; give **every** such $c$ ($\sin^2 x$ on $[0,\pi]$: $c = \frac{\pi}{4}, \frac{3\pi}{4}$).

Parametric $y_{avg} = \frac{1}{x(\beta) - x(\alpha)}\int_\alpha^\beta y(t)\,x'(t)\,dt$.

| See this | Do this |
|---|---|
| odd $f$ on $[-a,a]$ ($\tan^3 x \sec^4 x$, $\sin x$) | average is 0, no integral |
| $\sin^2, \cos^2$ | half-angle |
| odd power $\sin^3, \cos^5$ | peel one off, rest via $\sin^2 + \cos^2 = 1$, $u = \cos$ or $\sin$ |
| $\sec^4 x$ | $\sec^2 x(1 + \tan^2 x)$, $u = \tan x$ |
| $\frac{6}{x^2-1}$ / $\frac{16}{(x^2+4)^{3/2}}$ / $xe^{-x}$ | partial fractions / trig sub (§7) / parts |

$\overline{fg} \ne \bar{f}\,\bar{g}$. $\bar{f} < \bar{g}$ does not mean $f \le g$ everywhere.

## 4 · Centroids and center of mass

Top $f$, bottom $g$, $a \le x \le b$, density $\rho(x)$ (centroid: $\rho = 1$, $m = A$):

$m = \int_a^b \rho\,(f-g)\,dx \quad M_y = \int_a^b x\,\rho\,(f-g)\,dx \to \bar{x} = \frac{M_y}{m}$

$M_x = \frac{1}{2}\int_a^b \rho\,(f^2 - g^2)\,dx \to \bar{y} = \frac{M_x}{m}$ $\quad f^2 - g^2$**, not** $(f-g)^2$

Rod: $\bar{x} = \frac{\int x\rho\,dx}{\int \rho\,dx}$. Point masses: $\bar{x} = \frac{\sum m_i x_i}{\sum m_i}$.

Solid about x-axis, $\rho(x)$: $\bar{x} = \frac{\int x\rho\,\pi(R^2 - r^2)dx}{\int \rho\,\pi(R^2 - r^2)dx}$.

| See this | Do this |
|---|---|
| symmetric about $x = c$ (and $\rho$ symmetric too) | $\bar{x} = c$ free. About x-axis ⇒ $\bar{y} = 0$; about $y = x$ ⇒ $\bar{x} = \bar{y}$ |
| rectangles, triangles, circle pieces | composite $\bar{x} = \frac{\sum A_i \bar{x}_i}{\sum A_i}$; hole = negative area |
| "greater / less / equal" with a density | $\rho$ increasing ⇒ $\bar{x}$ moves right; decreasing ⇒ left; even/constant ⇒ no change |
| curves cross | solve $f = g$ for bounds; plug a middle point to see which is on top |
| $\ln x$ region | parts: $\int \ln x$, $\int x\ln x$, $\int (\ln x)^2$ (§7) |

| Shape | Centroid | Area |
|---|---|---|
| Rectangle $w \times h$ at origin | $(\frac{w}{2}, \frac{h}{2})$ | $wh$ |
| Upper semicircle, radius $r$ | $(0, \frac{4r}{3\pi})$ | $\frac{1}{2}\pi r^2$ |
| Quarter circle (1st quadrant) | $(\frac{4r}{3\pi}, \frac{4r}{3\pi})$ | $\frac{1}{4}\pi r^2$ |
| Any triangle | average of the 3 vertices | $\frac{1}{2}bh$ |

**Example.** $y = 4 - x^2$ over x-axis: symmetric ⇒ $\bar{x} = 0$; $A = \frac{32}{3}$, $M_x = \frac{1}{2}\int_{-2}^2 (4 - x^2)^2 dx = \frac{256}{15}$ ⇒ $\bar{y} = \frac{8}{5}$.

# MATH 1400 · Quiz 5 · Side 2 — Probability · Integration moves · Checks · Mistakes

Week 11 builds on Weeks 7–9: most pdf integrals are u-sub, parts or trig sub

## 5 · Probability density functions

Valid pdf: $f(x) \geq 0$ and $\int_{-\infty}^{\infty} f(x)\,dx = 1$. $P(a \leq X \leq b) = \int_a^b f(x)\,dx$, $P(X = a) = 0$.

Mean $\mu = \int x\,f(x)\,dx$. Median $m$: $\int_{\text{left end}}^{m} f(x)\,dx = \frac{1}{2}$.

$\sigma^2 = \int x^2 f(x)\,dx - \mu^2$ (subtract $\mu^2$!) $\sigma = \sqrt{\sigma^2}$.

CDF $F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt$, $f = F'$. $P(X > a) = 1 - F(a)$. Integrate only over the support.

| See this | Do this |
|---|---|
| "find $C$, $c$, $\alpha$ or $A$ so $f$ is a pdf" | set $\int_{\text{support}} f = 1$ and solve; the unknown can be a bound: $\int_1^{\alpha} x^{-1/2}dx = 1 \Rightarrow \alpha = \frac{9}{4}$ |
| given $P(X < x)$ as a formula | differentiate: $f = \frac{d}{dx}P(X < x)$. Given $P(X > a)$: $f = -\frac{d}{da}P(X > a)$ |
| $ke^{-kx}$ or $\frac{1}{\mu}e^{-x/\mu}$, $x \geq 0$ (exponential) | $\mu = \sigma = \frac{1}{k}$; median $\mu \ln 2$; $P(X > x) = e^{-x/\mu}$ |
| $x^n e^{-kx}$ on $[0, \infty)$ | $\int_0^{\infty} x^n e^{-kx}dx = \frac{n!}{k^{n+1}}$ ( $Cxe^{-x/2} \Rightarrow C = \frac{1}{4}$) |
| $2xe^{-x^2}$ | $u = x^2$: $P(X > k) = e^{-k^2}$ |
| $3x^2$, $4x^3$, $3(1-x)^2$ on $[0,1]$ | $F(m) = m^3$, $m^4$, $1-(1-m)^3 = \frac{1}{2}$ |
| $3\sin 3x$ on $[0, \frac{\pi}{6}]$ | $1 - \cos 3m = \frac{1}{2} \Rightarrow m = \frac{\pi}{9}$ |
| $\frac{1}{\sqrt{a^2-x^2}}$, $\frac{x^2}{\sqrt{1-x^2}}$, $\frac{1}{(a^2-x^2)^{3/2}}$ | arcsin, or $x = a\sin\theta$ (§7) |
| $\frac{1}{1+x^2}$ / $\frac{x}{1+x^2}$ / $\frac{x^2}{1+x^2}$ | arctan / $\frac{1}{2}\ln(1+x^2)$ / $1 - \frac{1}{1+x^2}$ |
| $\frac{x}{(1+x)^n}$ | $u = 1 + x$, $x = u - 1$ |
| $\frac{2\ln x}{x}$: $\int f$ / $\int xf$ | $u = \ln x$ / parts ($\int \ln x$) |
| $\frac{3}{x^4}$ on $[1, \infty)$ | power rule, $\lim_{b\to\infty}$; $\mu = \frac{3}{2}$, $\sigma^2 = 3 - \frac{9}{4}$ |
| uniform on $[a, b]$ | $f = \frac{1}{b-a}$, $\mu = \frac{a+b}{2}$, $\sigma^2 = \frac{(b-a)^2}{12}$ |
| normal | $z = \frac{x-\mu}{\sigma}$; symmetric, mean = median = $\mu$ |
| discrete values $x_i$ with $p_i$ | $E = \sum x_i p_i$, $\text{Var} = \sum (x_i - \mu)^2 p_i$ |

**Example.** $f = \frac{3}{4}(2x - x^2)$ on $[0,2]$: $\mu = \frac{3}{4}\int_0^2 (2x^2 - x^3)dx = 1$; $\int x^2 f = \frac{6}{5}$; $\sigma^2 = \frac{6}{5} - 1^2 = \frac{1}{5}$.

## 6 · Geometric probability

Uniform point in region $R$: $P(A) = \dfrac{\text{Area}(A)}{\text{Area}(R)}$. Complement $1 - P$ is often easier.

| See this | Do this |
|---|---|
| circles with a shared center | ratio of $r^2$; π cancels. $P(r > 3)$ in radius 5: $\frac{25-9}{25} = \frac{16}{25}$ |
| "within $\theta$ degrees" | sector: total angle / 360° ( $\pm 5^\circ \Rightarrow \frac{10}{360} = \frac{1}{36}$) |
| region under / between curves inside a rectangle | $\int(\text{top} - \text{bottom}) \div$ rectangle area; $x = y^2$ to $x = 9$: integrate in $y$ |
| line cuts a triangle or rectangle | small triangle $\frac{1}{2}bh$; find $b$ from the line equation |
| two arrival times in $[0, T]$, wait $w$ | $P(\text{meet}) = 1 - \frac{(T-w)^2}{T^2}$ |
| distance $R$ from center of a disk of radius $a$ | $F(r) = \frac{r^2}{a^2}$, $f = \frac{2r}{a^2}$ |

## 7 · Integration moves (from Weeks 7–9)

**u-sub:** change the bounds, or substitute back. If an $x$ is left over, write it in $u$ ($x = u - 1$).

**Parts:** $\int u\,dv = uv - \int v\,du$ (u = log, then polynomial).

$\int xe^{-x}dx = -xe^{-x} - e^{-x}$ $\int \ln x\,dx = x\ln x - x$

$\int x\ln x\,dx = \frac{x^2}{2}\ln x - \frac{x^2}{4}$ $\int (\ln x)^2 dx = x(\ln x)^2 - 2x\ln x + 2x$

**Trig sub results:**

$\int \frac{dx}{(x^2+a^2)^{3/2}} = \frac{x}{a^2\sqrt{x^2+a^2}}$

$\int \frac{dx}{(a^2-x^2)^{3/2}} = \frac{x}{a^2\sqrt{a^2-x^2}}$

$\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin\frac{x}{a}$ $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan\frac{x}{a}$

**Partial fractions:** $\frac{1}{x^2-a^2} = \frac{1}{2a}\left[\frac{1}{x-a} - \frac{1}{x+a}\right]$

**Trig:** $\sin^2 x = \frac{1-\cos 2x}{2}$, $\cos^2 x = \frac{1+\cos 2x}{2}$, $1 + \tan^2 = \sec^2$, $\int \sec x\,dx = \ln|\sec x + \tan x|$, $\cosh^2 - \sinh^2 = 1$

**Improper:** $\int_1^{\infty} x^{-p}dx = \frac{1}{p-1}$ ($p > 1$); write $\lim_{b\to\infty}$, $e^{-b} \to 0$.

## 8 · Answer shapes (check yourself)

- Arc lengths here come out clean (6, $\frac{14}{3}$, $\frac{21}{16}$, 147) or as $\ln(\dots)$. A messy root means you missed the perfect square.
- Surface area has a π and is positive.
- Centroid sits on every symmetry line; semicircle $\bar{y} \approx 0.42r$
- $f_{avg}$ is between the min and max of $f$.
- Probabilities are in $[0,1]$; $\mu$ and $m$ are inside the support; $\sigma^2 > 0$; exponential median $<$ mean.

## 9 · Workflow (every problem)

1. Pick the tool (§0).
2. Sketch: curve or region, axis, bounds (solve for intersections).
3. Pick $dx$, $dy$ or $dt$ from how the curve is given.
4. Write the full integral, with bounds and $dx$, before simplifying.
5. Simplify: perfect square, cancel the radius, use symmetry.
6. Integrate (§7); change the bounds with $u$.
7. Divide (average, centroid) or set equal to 1 or ½ (pdf, median).
8. Check against §8.

## 10 · Common mistakes

- **Surface area: $2\pi r\,dL$, not $2\pi r\,dx$.**
- **Variance: subtract $\mu^2$.**
- **$M_x = \frac{1}{2}\int(f^2 - g^2)$, not $\frac{1}{2}\int(f-g)^2$.**
- Forgetting to change the bounds after a substitution.
- Leaving $y$ in a $dx$ integral: the radius must be in the integration variable.
- $\int u^{-2}du = -\frac{1}{u}$ (minus sign).
- Mixing up $M_y \to \bar{x}$ and $M_x \to \bar{y}$.
- Median: set the integral of $f$ equal to ½, not $f$ itself.
- Integrating a pdf outside where it is nonzero.
- MVT: listing only one $c$ when there are two.

**My own mistakes** (add from graded work):